

Patient Shortage in Histopathology: Does Full-Feature Redundancy Explain the Residual Breadth Benefit?

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Abstract

At a fixed patch budget, histopathology patch classifiers trained on more patients are more accurate. On TCGA-UT, adjusting patch counts for within-patient redundancy measured along a single principal feature direction leaves a residual benefit of about three percentage points per doubling of patients. This experiment asks whether dependence across other feature directions explains that residual. A full-feature intraclass correlation replaces the single-direction measure on the same patient and patch samples, while predictions from the nine-cell breadth–depth grid remain fixed. Mean correlation increases from 0.457 to 0.487, and the residual breadth benefit falls from 2.98 to 2.62 percentage points per doubling (test-patient 95% interval: 2.04 to 3.17). Effective support alone fits the grid better, with residual standard deviation falling from 0.253 to 0.201 points. Nevertheless, the remaining breadth benefit exceeds the one-point practical threshold under both test-patient uncertainty and training-patient sensitivity. Full-feature redundancy explains only a small part of the residual breadth benefit: about 12% of the original estimate. Most of the benefit of additional patients remains unexplained by this correlation-based adjustment.

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1 Motivation

Patches from one patient share acquisition conditions and much of their morphology, so they are worth fewer independent observations than their number suggests. *Effective support* expresses this by dividing the patch count by a design effect that grows with the *intraclass correlation*, the share of feature variance that lies between patients rather than within them. The breadth–depth comparison of Queisler (2026) tested whether this adjustment explains why patient count matters. It trained multinomial logistic regression on frozen Virchow2 features using allocations that vary the number of patients and the patches per patient independently. At 160 patches per class, twenty patients contributing eight patches each outperformed five patients contributing 32 patches each by 9.52 percentage points on TCGA-UT (95% interval: 8.96 to 10.07). Even at equal effective support, each doubling of patients still added 2.98 points (2.47 to 3.46).

That residual need not indicate missing patient coverage. The correlation was computed on a scalar score, the projection of each feature vector onto the leading principal direction fitted across all 30 cancer types. TCGA-UT patches are rotated and rescaled crops of a few pathologist-selected uniform tumour regions per slide (Komura et al., 2022), so patches of one patient share tissue in every feature direction, not only along a direction that need not align with the variation within any single class. The scalar correlation may therefore understate redundancy. The breadth–depth grid cannot rule this out: correlations about 1.56 times as large as estimated would absorb the residual and fit the nine allocations equally well (Queisler, 2026).

The question is therefore: *does a correlation that uses all feature directions account for the residual breadth benefit on TCGA-UT?* If redundancy was underestimated, effective support computed from the full features should absorb the residual. If the residual persists, this full-feature adjustment is insufficient to account for the observed breadth benefit.

2 Study design

Only the redundancy measure changes. The design inherits the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features, three fixed patient-disjoint train–validation–test splits, and the nine allocations of Queisler (2026): 5, 10, or 20 patients per class, each contributing 8, 16, or 32 patches. TCGA-UT participants are called patients throughout. The five training draws per allocation, the logistic-regression predictions on the natural test partitions, and hence every cell accuracy stay fixed, so any change in the analysis is attributable to the correlation alone. Table 1 contrasts the two measures.

	Single-direction correlation	Full-feature correlation
Input	Projection onto the leading principal direction	All 2,560 feature coordinates
Dependence captured	Between-patient differences along one direction	Between-patient differences in total feature variance

Table 1: The two redundancy measures differ only in which feature variation they use. Both are computed from the same sampled patients and patches, both apply the same correction for unequal patient sizes, and both are evaluated against the same cell accuracies.

3 Full-feature correlation

The full-feature correlation follows the covariance-trace form of multivariate intraclass correlation (Shou et al., 2013): variance is summed over all coordinates first, and the correlation is

then formed from these totals. It is applied here to patch embeddings with patients as clusters, and serves as a descriptive measure of redundancy rather than as a validated effective sample size for classifier learning.

For one class in one training split, let H patients contribute n_i feature vectors x_{ij} , with $N = \sum_i n_i$ patches in total. Let $\bar{x}_i$ denote the mean feature vector of patient i and $\bar{x}$ the mean over all patches. Four quantities define the estimator. The *between-patient mean square* B measures how far patient means lie from the overall mean, weighting each patient by its number of patches. The *within-patient mean square* W measures how far patches lie from their own patient’s mean. Because patients contribute unequal numbers of patches, the *effective patient size* $\tilde{m}$ replaces the common patch count of a balanced design. Together they give the correlation $\hat{\rho}_{\text{full}}$:

$$\begin{aligned} B &= \frac{\sum_i n_i \|\bar{x}_i - \bar{x}\|^2}{H - 1}, & W &= \frac{\sum_i \sum_j \|x_{ij} - \bar{x}_i\|^2}{N - H}, \\ \tilde{m} &= \frac{N - \sum_i n_i^2 / N}{H - 1}, & \hat{\rho}_{\text{full}} &= \frac{B - W}{B + (\tilde{m} - 1)W}. \end{aligned} \quad (1)$$

For scalar scores, the squared norms reduce to squared differences and Equation (1) becomes the one-way random-effects estimator with unequal-cluster correction used for the single-direction correlation (Kenny and La Voie, 1985). The two measures thus differ only in their input.

The estimator reads intuitively. If a patient’s patches were independent draws around the class mean, patient means would scatter only through sampling noise, B would be close to W , and the correlation would be close to zero. If a patient’s patches were near-copies of each other, W would be small relative to B , and the correlation would approach one. Because each squared norm adds squared differences over all 2,560 coordinates, B and W collect patient-specific shifts from every feature direction, so a patient whose patches share tissue in directions other than the leading principal direction still raises the correlation. Directions with larger variance carry more weight. Rotating the feature basis leaves the estimate unchanged, whereas rescaling individual coordinates does not; the features therefore keep their original scale, without whitening or selection of directions.

Both measures use the same sample: for every class and training split, at most 200 patients and at most 200 patches per patient, drawn from the full training partition as in Queisler (2026). The two correlations therefore differ only in how dependence is summarized, not in which material it is measured on. Estimates are clipped to $[0, 1]$ before use, as for the single-direction measure, and raw values are reported. An estimate is undefined when fewer than two patients or no within-patient degrees of freedom are available, or when the denominator is not positive; such estimates are neither set to zero nor omitted.

For each measure, the clipped correlations of a class are averaged over the three splits to give $\bar{\rho}_c$. With G patients per class each contributing m patches, effective support of class c is

$$N_{\text{eff},c}(G, m) = \frac{Gm}{1 + (m - 1)\bar{\rho}_c}, \quad (2)$$

and the cell predictor $N_{\text{eff}}(G, m)$ averages $N_{\text{eff},c}$ over classes. Larger correlations reduce effective support most at large depth, where many patches come from each patient, so a larger full-feature correlation lowers the deep cells more than the broad ones.

4 Evaluation

The analysis asks whether a better measure of redundancy leaves less for patient count to explain. It refits the support regression of Queisler (2026) once with each correlation measure and compares how much breadth benefit remains.

4.1 Residual breadth benefit

Accuracy is patient-macro balanced accuracy: patch recall averaged first within each test patient and true class, then equally across patients and classes. A cell accuracy $A(G, m)$, in percent, averages the three splits and five draws equally. Across the nine cells, the regression

$$A(G, m) = \alpha + \beta \log N_{\text{eff}}(G, m) + \gamma \log G, \quad b = \gamma \log 2, \quad (3)$$

separates two contributions. The term $\beta \log N_{\text{eff}}$ captures what effective support explains, and $\gamma \log G$ captures what patient count adds beyond it. The *residual breadth benefit* b is the resulting gain in percentage points per doubling of patients at fixed effective support. With the single-direction correlation, b is 2.98 points (Queisler, 2026). If that correlation understated redundancy, the full-feature measure should lower the effective support of deep cells, attribute more of the breadth benefit to $\beta \log N_{\text{eff}}$, and shrink b . The regression is a descriptive summary of the nine cells, not a matched-support intervention.

A smaller b alone is not sufficient, because a correlation can shift the breadth benefit away from γ while making effective support a worse description of the grid. The analysis therefore also fits effective support alone, $A = \alpha + \beta \log N_{\text{eff}}$, and compares its residual standard deviation, with seven residual degrees of freedom, between the two measures. The full-feature measure counts as fitting no worse when this value does not exceed that of the single-direction measure, which was 0.25 points (Queisler, 2026). Neither correlation is adjusted to improve the fit. Class correlations and cell effective support are reported alongside, showing where the two measures disagree.

4.2 Uncertainty

Two sources of uncertainty are reported separately, because they answer different questions.

Test-patient uncertainty asks how much the result depends on which patients happen to form the test partitions. Each of the 1,999 paired Bayesian bootstrap draws of Queisler (2026) assigns Dirichlet weights to the distinct test patients (Rubin, 1981). A patient’s weight is shared across both measures, all cells, draws, splits, classes, and repeated test appearances, so differences between the measures are not blurred by different resamples. Accuracies and regressions are recomputed with correlations held fixed, and the 2.5th and 97.5th percentiles give exploratory 95% intervals.

Training-patient sensitivity asks how much the result depends on which training patients entered the correlation estimates. A separate set of 1,999 cluster bootstrap draws resamples the distinct patients in the correlation samples with replacement. A resampled patient keeps all of its sampled patches in every class and split in which it appears, and both measures use the same resample. Correlations, effective support, and regressions are recomputed, while cell accuracies stay fixed. These intervals are not combined with the test-patient intervals and do not describe retraining on new cohorts. Each stored distribution contains 2,000 entries: the observed estimate followed by 1,999 resampled estimates; percentile intervals exclude the observed entry. Accuracy dispersion across the 15 split–draw combinations accompanies both analyses. It includes variation between splits and is not a pure within-split measure of training-draw variability.

5 Interpretation criteria

Conclusions rest on the test-patient interval for the full-feature residual b and on the practical threshold of one percentage point per doubling of patients (Queisler, 2026). Training-patient sensitivity accompanies each conclusion; a conclusion that changes under it is reported as fragile.

Underestimated redundancy. If the interval for b lies within $[-1, 1]$ points and effective support alone fits no worse than with the single-direction correlation, the result supports the account that the leading principal direction missed relevant redundancy. Full-feature effective support would then be the more complete summary of patient shortage on TCGA-UT.

Breadth beyond redundancy. If the interval lies above one point, additional patients retain a practically relevant benefit that this redundancy measure does not explain. Patients would then contribute variation beyond their effective observations, such as additional morphology or institution-specific acquisition; the present design does not identify which.

Inconclusive outcomes. All other outcomes, including an interval that crosses a threshold or a small residual accompanied by a worse fit of effective support alone, leave both accounts open.

Neither outcome establishes a causal mechanism. Both measures compress dependence into one scalar per class and assume that patches within a patient are exchangeable. All results are conditional on Virchow2 features, logistic regression, the sampled training patients, and the range of the nine allocations. Interventions on institutions, comparisons of representations, and methods that mitigate patient shortage lie outside the scope of this experiment.

6 Results

Full-feature correlation reduces the residual breadth benefit but does not bring it below practical relevance. The analysis retains all 30 cancer types, three splits, and nine allocations, each averaging five training draws per split. The 135 stored classifier evaluations are unchanged. Both correlation measures are defined for all 90 class-split combinations and for every training bootstrap draw; all raw correlations lie within $[0, 1]$, so clipping has no effect on these results.

6.1 Residual breadth and quality of fit

The full-feature residual is 2.62 percentage points per doubling of patients, compared with 2.98 for the single-direction measure (Table 2). Its test-patient interval, 2.04 to 3.17 points, lies entirely above the one-point practical threshold. The training-patient sensitivity interval, 2.60 to 2.70 points, gives the same interpretation. Under the criteria in Section 5, the outcome therefore supports breadth beyond the redundancy captured by this measure, and the classification does not change under training-patient resampling.

Quantity	Measure	Estimate	Test-patient interval	Training-patient interval
Residual breadth	Single direction	2.98	[2.47, 3.46]	[2.93, 3.21]
	Full features	2.62	[2.04, 3.17]	[2.60, 2.70]
	Full minus single	-0.36	[-0.44, -0.28]	[-0.53, -0.30]
Fit error	Single direction	0.253	[0.207, 0.398]	[0.245, 0.298]
	Full features	0.201	[0.151, 0.365]	[0.199, 0.212]
	Full minus single	-0.051	[-0.059, -0.030]	[-0.090, -0.043]

Table 2: Full-feature correlation reduces both the residual breadth benefit and the error of the effective-support-only fit. Residual breadth is in percentage points per doubling of patients; fit error is the residual standard deviation in percentage points across nine cells, with seven residual degrees of freedom. Test-patient and training-patient 95% intervals describe separate sources of uncertainty. Differences use paired draws, not differences between interval endpoints.

The paired reduction in residual breadth is 0.36 points (test-patient interval for full minus single: -0.44 to -0.28 ; training-patient interval: -0.53 to -0.30), or approximately 12% of the original

point estimate. This reduction is accompanied by a better fit of effective support alone: residual standard deviation decreases from 0.253 to 0.201 points. The paired change is -0.051 points, with both uncertainty intervals below zero. Thus, the failure to absorb the breadth benefit cannot be attributed to a deterioration in this fit criterion.

6.2 Correlations, support, and unchanged accuracy

The class-averaged correlation increases from 0.457 to 0.487. Full-feature estimates exceed single-direction estimates in 22 of 30 classes, but the change is not uniform (Appendix A). The largest increase is for uveal melanoma, from 0.329 to 0.476, whereas kidney renal clear cell carcinoma decreases from 0.641 to 0.548. Across classes, single-direction estimates range from 0.314 to 0.641 and full-feature estimates from 0.367 to 0.582. Using all directions therefore changes the distribution of class correlations rather than simply multiplying them by a common factor.

G	m	Accuracy [95% interval]	SD	N_{eff} single	N_{eff} full
5	8	61.97 [60.73, 63.09]	1.31	9.73	9.15
5	16	62.13 [60.90, 63.23]	1.50	10.47	9.74
5	32	62.25 [61.02, 63.34]	1.54	10.90	10.06
10	8	67.04 [65.78, 68.25]	1.13	19.46	18.29
10	16	67.28 [66.01, 68.51]	1.14	20.95	19.47
10	32	67.39 [66.10, 68.58]	1.11	21.79	20.12
20	8	71.76 [70.43, 72.94]	0.54	38.92	36.58
20	16	72.02 [70.68, 73.20]	0.53	41.89	38.94
20	32	72.08 [70.75, 73.26]	0.52	43.58	40.25

Table 3: The nine allocations retain identical accuracies under both redundancy measures. G is patients per class and m is patches per patient. Accuracy is patient-macro balanced accuracy in percent, with test-patient 95% intervals. SD is the population standard deviation across all 15 split-draw accuracies, in percentage points. Effective support is averaged over classes and expressed per class.

Full-feature effective support is lower in every cell (Table 3). The relative reduction grows with depth: approximately 6.0% at eight patches per patient, 7.1% at sixteen, and 7.7% at thirty-two. At the fixed budget of 160 patches per class, effective support falls from 10.90 to 10.06 for five patients contributing 32 patches each, and from 38.92 to 36.58 for twenty patients contributing eight patches each. The corresponding accuracy difference remains 9.52 percentage points: changing the redundancy measure changes the explanation of the grid, not the predictions.

Increasing depth from eight to thirty-two patches improves accuracy by only 0.28 to 0.34 points at fixed breadth. Increasing breadth from five to twenty patients improves accuracy by 9.80 to 9.88 points at fixed depth. The full-feature adjustment reflects this weak depth response more closely, but does not fully explain the breadth response. Dispersion across the 15 split-draw evaluations ranges from 0.52 to 1.54 points; these descriptive values include split variation and should not be read as uncertainty intervals for the fitted residual.

7 Discussion and conclusion

Full-feature redundancy explains only a small part of the residual breadth benefit on TCGA-UT. Accounting for dependence across all feature directions lowers the residual by about 12%, while 2.62 percentage points per doubling of patients remain. Together with the improved fit of effective support alone, this suggests that the single-direction measure missed some relevant dependence, but that measuring total feature redundancy more completely is insufficient to explain why additional patients help.

The contrast between breadth and depth points towards a distinction between repeated observations and coverage of different patient characteristics. Quadrupling the patches per patient adds at most 0.34 points, whereas quadrupling patient count adds approximately 9.8 points. One hypothesis is that extra patches mainly refine patterns already represented by a patient’s tissue, while extra patients introduce patterns absent from the smaller cohort. Effective support describes how much observations repeat each other, but does not directly describe which diagnostically relevant patterns the training set covers. This could explain why it follows the overall accuracy trend closely yet leaves a systematic breadth benefit.

A concrete candidate is variation associated with the institutions contributing TCGA-UT slides. Additional patients may broaden the range of staining, preparation, or scanning conditions encountered during training, making cancer recognition less dependent on a narrow set of acquisition conditions. Another candidate is biological variation within each cancer type: additional patients may contribute morphological subtypes or tumour appearances that repeated crops from the same tissue cannot supply. Both hypotheses predict a benefit from covering distinct patient characteristics, rather than from increasing the number of observations alone. The present result motivates these explanations without distinguishing between them.

The class-specific changes also suggest that the missing information may depend on the classification task. Full-feature correlation increases in 22 classes but decreases in eight, showing that the leading direction does not uniformly underestimate within-patient dependence. Moreover, total feature variance need not emphasize the directions that distinguish one cancer type from another. A patient could therefore add useful variation near a class boundary without greatly changing the covariance-trace correlation. A useful shortage signal would need to capture this contribution to class coverage, rather than assume that all feature variation has equal predictive value.

A follow-up experiment could test institutional coverage directly. For cancer types with sufficient source-site metadata and patient counts, training cohorts would vary the number of represented institutions while holding patients and patches per class fixed, with evaluation on a common patient-disjoint test set. Repeating this comparison across patient counts would test whether institutional coverage explains part of the remaining breadth benefit. A positive result would motivate a measurable institution-coverage signal and, subsequently, a mitigation method targeting that shortage.

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A Class-specific correlation estimates

Table 4 reports the split-averaged correlations used to calculate effective support and their training-patient sensitivity intervals. These are descriptive class-level comparisons; no adjustment for multiple class comparisons is applied. Table 5 gives the corresponding raw estimates for each split. All estimates, including the resampled values, lie within $[0, 1]$, so raw and clipped point estimates coincide.

Cancer type	Single direction	Full features
Adrenocortical carcinoma	0.500 [0.315, 0.620]	0.433 [0.381, 0.473]
Bladder Urothelial Carcinoma	0.508 [0.462, 0.549]	0.521 [0.504, 0.537]
Brain Lower Grade Glioma	0.429 [0.376, 0.477]	0.484 [0.465, 0.498]
Breast invasive carcinoma	0.448 [0.402, 0.489]	0.511 [0.497, 0.523]
Cervical squamous cell / endocervical adenocarcinoma	0.515 [0.456, 0.566]	0.516 [0.495, 0.533]
Colon adenocarcinoma	0.394 [0.310, 0.475]	0.425 [0.395, 0.450]
Esophageal carcinoma	0.410 [0.328, 0.481]	0.518 [0.490, 0.538]
Glioblastoma multiforme	0.337 [0.277, 0.393]	0.400 [0.380, 0.418]
Head and neck squamous cell carcinoma	0.329 [0.283, 0.373]	0.383 [0.364, 0.400]
Kidney Chromophobe	0.489 [0.365, 0.587]	0.553 [0.512, 0.584]
Kidney renal clear cell carcinoma	0.641 [0.581, 0.689]	0.548 [0.522, 0.571]
Kidney renal papillary cell carcinoma	0.564 [0.503, 0.612]	0.555 [0.535, 0.571]
Liver hepatocellular carcinoma	0.521 [0.468, 0.569]	0.532 [0.513, 0.549]
Lung adenocarcinoma	0.398 [0.360, 0.433]	0.442 [0.426, 0.457]
Lung squamous cell carcinoma	0.314 [0.275, 0.350]	0.367 [0.353, 0.378]
Mesothelioma	0.414 [0.279, 0.519]	0.506 [0.457, 0.539]
Ovarian serous cystadenocarcinoma	0.513 [0.404, 0.599]	0.467 [0.428, 0.495]
Pancreatic adenocarcinoma	0.328 [0.269, 0.383]	0.434 [0.398, 0.464]
Pheochromocytoma and Paraganglioma	0.636 [0.491, 0.721]	0.582 [0.523, 0.619]
Prostate adenocarcinoma	0.527 [0.441, 0.600]	0.506 [0.476, 0.534]
Rectum adenocarcinoma	0.530 [0.393, 0.626]	0.520 [0.471, 0.560]
Sarcoma	0.493 [0.419, 0.552]	0.526 [0.500, 0.547]
Skin Cutaneous Melanoma	0.462 [0.410, 0.511]	0.542 [0.525, 0.555]
Stomach adenocarcinoma	0.452 [0.404, 0.494]	0.487 [0.468, 0.502]
Testicular Germ Cell Tumors	0.493 [0.429, 0.553]	0.500 [0.467, 0.530]
Thymoma	0.358 [0.256, 0.444]	0.477 [0.430, 0.512]
Thyroid carcinoma	0.520 [0.468, 0.564]	0.569 [0.551, 0.586]
Uterine Carcinosarcoma	0.378 [0.218, 0.544]	0.379 [0.330, 0.425]
Uterine Corpus Endometrial Carcinoma	0.485 [0.439, 0.524]	0.464 [0.448, 0.477]
Uveal Melanoma	0.329 [0.252, 0.390]	0.476 [0.440, 0.499]

Table 4: Class-specific correlations averaged over three training splits, with separate 95% training-patient sensitivity intervals for each measure. Full-feature correlation is higher in 22 of 30 classes; increases are not uniform across cancer types.

Cancer type	Single direction			Full features		
	Split 0	Split 1	Split 2	Split 0	Split 1	Split 2
Adrenocortical carcinoma	0.501	0.473	0.527	0.431	0.427	0.441
Bladder Urothelial Carcinoma	0.503	0.515	0.507	0.529	0.524	0.510
Brain Lower Grade Glioma	0.448	0.435	0.404	0.488	0.479	0.485
Breast invasive carcinoma	0.446	0.444	0.454	0.511	0.513	0.510
Cervical squamous cell / endocervical adenocarcinoma	0.518	0.512	0.516	0.516	0.518	0.514
Colon adenocarcinoma	0.379	0.405	0.397	0.421	0.434	0.421
Esophageal carcinoma	0.438	0.405	0.387	0.529	0.521	0.503
Glioblastoma multiforme	0.350	0.293	0.367	0.400	0.391	0.410
Head and neck squamous cell carcinoma	0.340	0.310	0.337	0.387	0.377	0.387
Kidney Chromophobe	0.490	0.523	0.454	0.553	0.558	0.549
Kidney renal clear cell carcinoma	0.639	0.653	0.630	0.551	0.553	0.540
Kidney renal papillary cell carcinoma	0.569	0.572	0.551	0.553	0.559	0.553
Liver hepatocellular carcinoma	0.516	0.515	0.531	0.531	0.528	0.538
Lung adenocarcinoma	0.405	0.394	0.394	0.439	0.447	0.440
Lung squamous cell carcinoma	0.291	0.329	0.323	0.360	0.369	0.372
Mesothelioma	0.391	0.435	0.417	0.501	0.520	0.495
Ovarian serous cystadenocarcinoma	0.544	0.479	0.516	0.482	0.454	0.465
Pancreatic adenocarcinoma	0.338	0.300	0.345	0.436	0.429	0.439
Pheochromocytoma and Paraganglioma	0.603	0.651	0.654	0.565	0.588	0.594
Prostate adenocarcinoma	0.538	0.528	0.516	0.505	0.504	0.508
Rectum adenocarcinoma	0.551	0.501	0.538	0.522	0.513	0.526
Sarcoma	0.502	0.497	0.478	0.528	0.529	0.520
Skin Cutaneous Melanoma	0.467	0.466	0.452	0.544	0.541	0.540
Stomach adenocarcinoma	0.478	0.436	0.441	0.493	0.483	0.484
Testicular Germ Cell Tumors	0.507	0.508	0.465	0.516	0.503	0.482
Thymoma	0.352	0.365	0.356	0.476	0.476	0.478
Thyroid carcinoma	0.526	0.508	0.527	0.572	0.568	0.567
Uterine Carcinosarcoma	0.376	0.342	0.415	0.387	0.370	0.381
Uterine Corpus Endometrial Carcinoma	0.458	0.504	0.493	0.460	0.467	0.466
Uveal Melanoma	0.314	0.327	0.344	0.461	0.490	0.476

Table 5: Unclipped correlations for each class and training split, computed on the same sampled patients and patches for both measures. Every estimate is finite and positive. Split variation is descriptive because the training partitions overlap.